

# Sentinel

## A Fully Procedural Game Engine and Open-World Military Shooter

Nothing is loaded from a model, image or sound file: everything is generated from code  
and a seed.

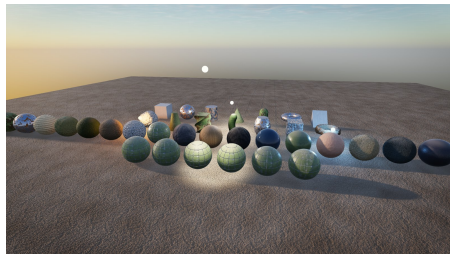
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# Outline

1. Problem, objectives, architecture
2. Procedural shapes and textures
3. The rendering pipeline
4. World, physics and gameplay
5. Results and verification
6. Challenge, limits, future work
7. Video tour of the project



Every material is a code-generated recipe.

# 1. Problem, objectives and architecture

**Can a complete, playable 3D game be made by code alone on a student laptop?**

## Two hard limits

- No model, image or precomputed data file
- macOS OpenGL 4.1: no compute shaders

## Objectives

- Layered engine, downward dependencies
- All content from code and seeds
- Modern renderer, playable game
- Verified: tests, 16.7 ms budget

Game: mission, AI, vehicles, sound

Render: deferred, lights, sky, post

World: collision, terrain

Procedural: noise, textures

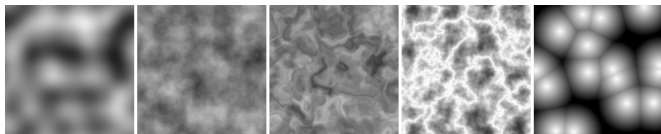
GPU: shaders, buffers

Platform: window, input



A lower layer never knows a higher one.

## 2. Procedural shapes and textures



gradient, fractal, warped, ridged and cellular noise

Fractal noise (tiles exactly)

$$F(\mathbf{x}) = \frac{\sum_{i=0}^{n-1} 2^{-i} v(2^i \mathbf{x})}{\sum_{i=0}^{n-1} 2^{-i}}$$

Vertex push

$$\mathbf{p}' = \mathbf{p} + A v(f\mathbf{p}) \hat{n}, \quad |\mathbf{p}' - \mathbf{p}| \leq A$$

- 18 recipes  $\rightarrow$  22 materials, baked once at  $1024^2$
- **Scharr** filter turns height into bumps; crevices darken
- Objects = tables of primitives + noise deformers
- Soldiers: skeletons with two-bone IK legs

### 3. The rendering pipeline



**Deferred shading:** store each pixel's colour, normal, shininess and depth first, then light it once.

#### Cook–Torrance reflection

$$f = \frac{(1-F_l)(1-F_v)c_d}{\pi} + DVF$$

$$D = \frac{\alpha^2}{\pi((\mathbf{n} \cdot \mathbf{h})^2(\alpha^2 - 1) + 1)^2}, F = F_0 + (1 - F_0)(1 - \mathbf{v} \cdot \mathbf{h})^5$$

#### Fog and tone map

$$C = TC_s + (1 - T)C_h, T = e^{-kd}$$

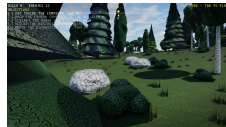
$$C_{\text{out}} = \frac{x(2.51x+0.03)}{x(2.43x+0.59)+0.14}$$



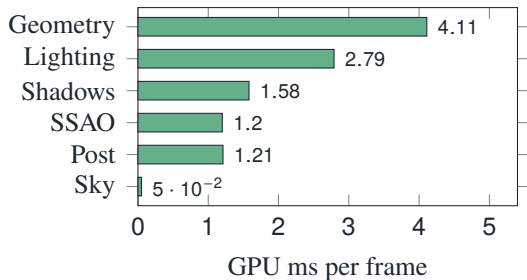
Dusk: sun, sky, haze, shadows

## 4. World, physics and gameplay

- **Sky:** sun height from astronomy; scattering; wind-driven clouds  
$$\mathbf{P}_{xz} = \frac{H}{d_y}(d_x, d_z) + \mathbf{w}t$$
- **Terrain:** warped ridged noise, streamed in 64 m chunks
- **Trees:** 4 species, swaying in the wind
- **Vehicles:** jeep, truck, carrier, tank, helicopter;  $\dot{\psi} = \frac{v \tan \delta}{L}$
- **Ladders:** 4-state grip machine
- **Mission:** hack, destroy, rescue, extract; stealth AI; synthesised sound



## 5. Results and verification



**10.9 ms** of the 16.7 ms budget at 1080p

|                              |         |
|------------------------------|---------|
| Unit tests (13 packages)     | 281     |
| TextureCheck (exact tiling)  | 8 318   |
| RenderCheck                  | 318 567 |
| GpuCheck                     | 30      |
| Playthroughs, 5/5 objectives | 2       |



## 6. Challenge solved, limits and future work

**Bug: a bright vertical line at night.** The fog colour sampled the *whole* sky, so one horizon star painted a streak on every surface in its column. Fix: fog uses the smooth atmosphere only.



before



after

### Limitations

Blocky primitive models; no true global illumination; lumpy foliage; tuned by scripts and screenshots.

### Future work

Leaf cards, screen-space GI, point-light shadows, multiplayer, a compute-shader port.

### Take-away

Most “art” is a few simple ideas, applied consistently and tested.

## 7. Video tour of the project

# References and thank you

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# Thank you

Questions are welcome



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